

Smart Traffic Lights using Local AI

Student Research Paper

Abstract— Smart traffic-light systems are being studied to reduce waiting times and adapt rapidly to dynamic road conditions. This paper provides a literature review comparing traditional fixed-timer traffic signals, cloud-based AI systems, and local edge-based smart cameras. We outline a standard four-step system architecture, evaluate system performance metrics, and highlight key research gaps regarding emergency vehicle prioritization.

Index Terms— Artificial Intelligence, Edge Computing, IEEE Style, Smart Traffic Lights, Traffic Signal Control.

I. INTRODUCTION

Traditional traffic lights usually follow fixed timers. While simple and inexpensive, they cannot easily react when a single road experiences sudden congestion. Modern adaptive systems leverage cameras and artificial intelligence (AI) to estimate vehicle density and dynamically adjust signal timing. Instead of relying strictly on clock cycles, decisions are informed by real-time roadway metrics. Recent reviews confirm a growing interest in AI-based traffic signal control and its potential benefits under changing conditions [1].

II. LITERATURE REVIEW

A. Cloud vs. Local Processing

Cloud-based AI systems transmit camera feed data to a central server for processing. Although this offers high computing capacity, it introduces dependency on stable internet connectivity and increases communication latency. Conversely, local smart-camera (edge computing) systems execute processing near the roadway. This minimizes bandwidth consumption and supports faster decision-making. Small local computers analyze localized camera feeds to issue immediate instructions to the signal controller. Machine-vision research highlights the efficiency of camera-based adaptive signal modes [2].

B. Critical Evaluation of Microcontroller Capabilities

Literature indicates extreme claims regarding low-cost microcontrollers. For example, Smith et al. [3] presented a claim where a basic Arduino Uno processed 10,000 live HD video streams simultaneously with zero delay. When evaluating such literature, it is essential to distinguish between actual hardware computational limits (such as 8-bit, 16 MHz hardware constraint boundaries) and theoretical or simulated claims.

III. SYSTEM ARCHITECTURE

The proposed system architecture is structured into four sequential steps:

- 1) Street Cameras: Capture live traffic conditions on-site.
- 2) Smart Pole Computers: Perform localized video processing and issue rapid traffic management decisions.
- 3) City Cloud Server: Receives selected summarized data for long-term storage and system-wide analytics.
- 4) Traffic Lights: Receive actuation decisions to adjust signal indicators.

The data and decision flow is summarized as follows:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

IV. PERFORMANCE COMPARISON

Table I summarizes the three primary traffic light system paradigms based on response time, bandwidth demand, and installation costs.

System Type	Decision Time	Internet Data	Setup Cost
Basic Timed Lights	30.0 s	Zero	\$200
Cloud AI Systems	2.0 s	High	\$2,500

System Type	Decision Time	Internet Data	Setup Cost
Local Smart Cameras	0.5 s	Low	\$1,200

TABLE I. COMPARISON OF TRAFFIC SYSTEM ARCHITECTURES

V. METHODOLOGY

The proposed smart traffic-light system uses a local AI-based architecture to analyze traffic conditions and make rapid signal-control decisions. The methodology consists of four main stages. First, street cameras capture real-time traffic conditions at different road intersections. Second, smart pole computers process the camera feeds locally using edge-based video processing to estimate vehicle density and identify changing traffic conditions. Third, selected summarized information is transmitted to the city cloud server for long-term storage and system-wide analysis. Finally, the traffic-light controller receives the decision generated by the local processing unit and adjusts the signal timing accordingly.

The local processing approach reduces dependency on continuous internet connectivity and minimizes communication latency. The system can therefore respond more quickly to sudden changes in traffic conditions compared with fixed-timer systems. The proposed architecture also provides a foundation for integrating emergency vehicle detection and priority-based traffic management in future implementations.

VI. RESEARCH GAP

Existing traffic signal control solutions function effectively under regular, predictable daily commuting patterns. However, a significant research gap persists in emergency scenarios: many existing systems lack integrated protocols to automatically clear paths for approaching emergency vehicles, such as speeding ambulances. There is a distinct need for smart traffic architectures capable of detecting priority vehicles and dynamically coordinating surrounding intersections to route them safely.

VII. FUTURE SCOPE

The proposed system can be extended to support automatic emergency vehicle prioritization. Future versions could use AI-based vehicle detection to identify approaching ambulances, fire engines, and other priority vehicles and provide them with a clear traffic path. Multiple intersections could also be coordinated so that traffic signals are adjusted along the emergency vehicle's route.

Further improvements may include advanced machine-vision techniques for more accurate vehicle-density estimation, improved coordination between local edge devices and the city cloud, and large-scale traffic analytics. These developments could help the system adapt to more complex traffic conditions while maintaining low response time and reduced bandwidth requirements.

VIII. CONCLUSION

Smart traffic lights using local AI provide an approach for adapting traffic signals to changing roadway conditions. Unlike traditional fixed-timer systems, the proposed architecture uses street cameras and local smart-pole computers to process traffic information and make rapid decisions. Local processing can reduce communication latency and bandwidth requirements while allowing selected information to be stored and analyzed through a city cloud server.

The comparison presented in this paper highlights the potential advantages of local smart-camera systems in terms of decision time and bandwidth demand. A key research gap remains the automatic prioritization of emergency vehicles. Addressing this gap can provide an important direction for future research in intelligent and responsive traffic signal control.

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